

Adding UA daily 4-km SWE

NSIDC-0719 v1 — University of Arizona gridded SWE + Snow Depth

A new source for **two** calibration targets: SWE and snow-covered area

Decision review before wiring aggregation + target builders. The fetch + consolidation layer is **done** (42 water years, 1982–2023, on the shared datastore); these are the consolidated, pre-projected source grids the aggregator would read next.

Broxton, P., Zeng, X. & Dawson, N. (2019). NSIDC-0719 v1, DOI [10.5067/0GGPB220EX6A](https://doi.org/10.5067/0GGPB220EX6A) . · Figures: [docs/figures/consolidated/gfv2-spatial-targets/](#)

What is NSIDC-0719, and why two targets?

Product	UA Daily 4-km Gridded SWE + Snow Depth, CONUS
Method	Assimilates SNOTEL / COOP in-situ snow obs with PRISM modeled fields (Broxton/Zeng/Dawson, U. Arizona)
Coverage	Water years 1982-2023 · daily · ~4 km · native NAD83 lat/lon
Variables	swe (kg m ⁻² ≡ mm) · snow_depth (mm)
In this repo	Pre-projected to EPSG:5070 at consolidate time (mirrors SNODAS — ~5-8× faster weight-gen) → <datastore>/ua_swe/daily/ua_swe_daily_WY<YYYY>.nc

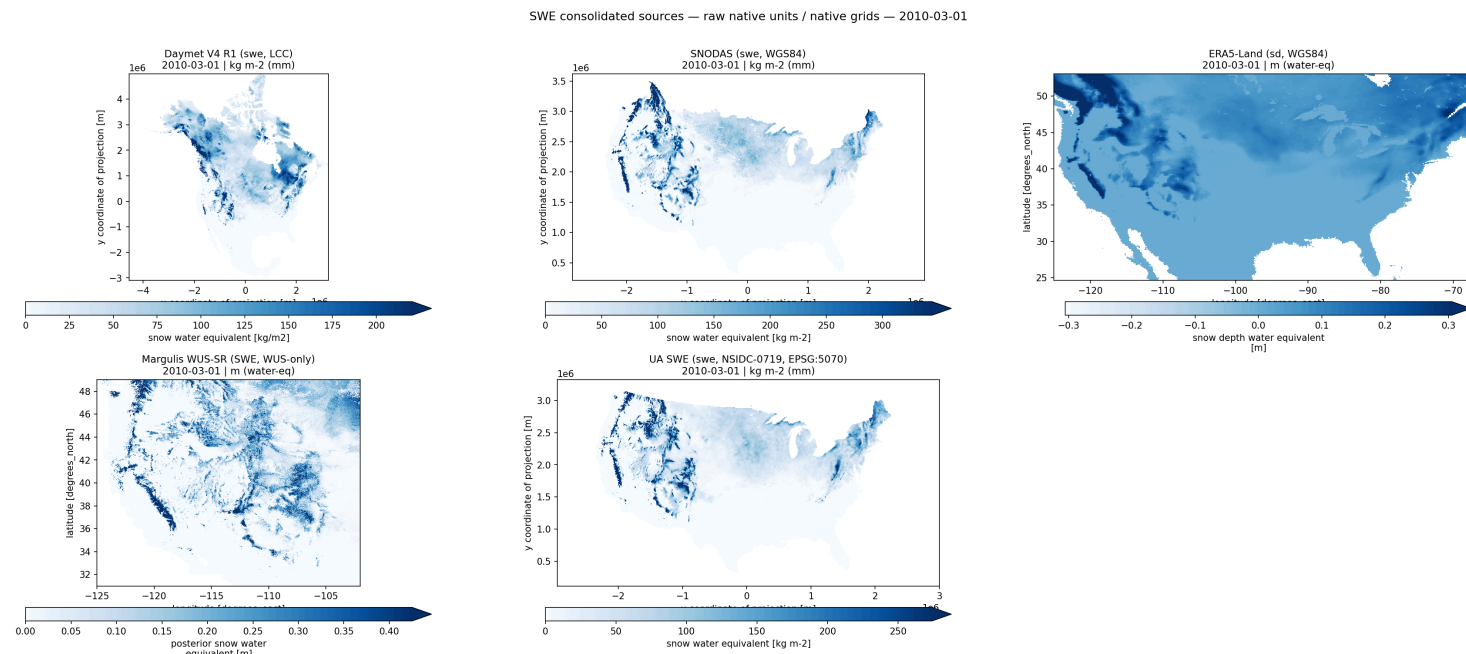
Value proposition #1 — a third CONUS SWE source, back to 1982.

Today's SWE bound leans on SNODAS (2003+) and Daymet; UA SWE adds an independent assimilation product and **pushes the multi-source record back two decades** before SNODAS exists.

Value proposition #2 — a second snow-covered-area source.

snow_depth > threshold is a per-pixel binary snow indicator; area-weighted to HRU it yields fractional SCA. This **breaks MOD10C1's single-source lock** on the SCA target and is **not weather-blocked** (MOD10C1 drops cloudy days).

SWE — native grids, peak winter (1 Mar 2010)



Five SWE sources at their native resolution / CRS. UA SWE (NSIDC-0719) is the new panel; the other four are the current SWE target inputs.

What to look for

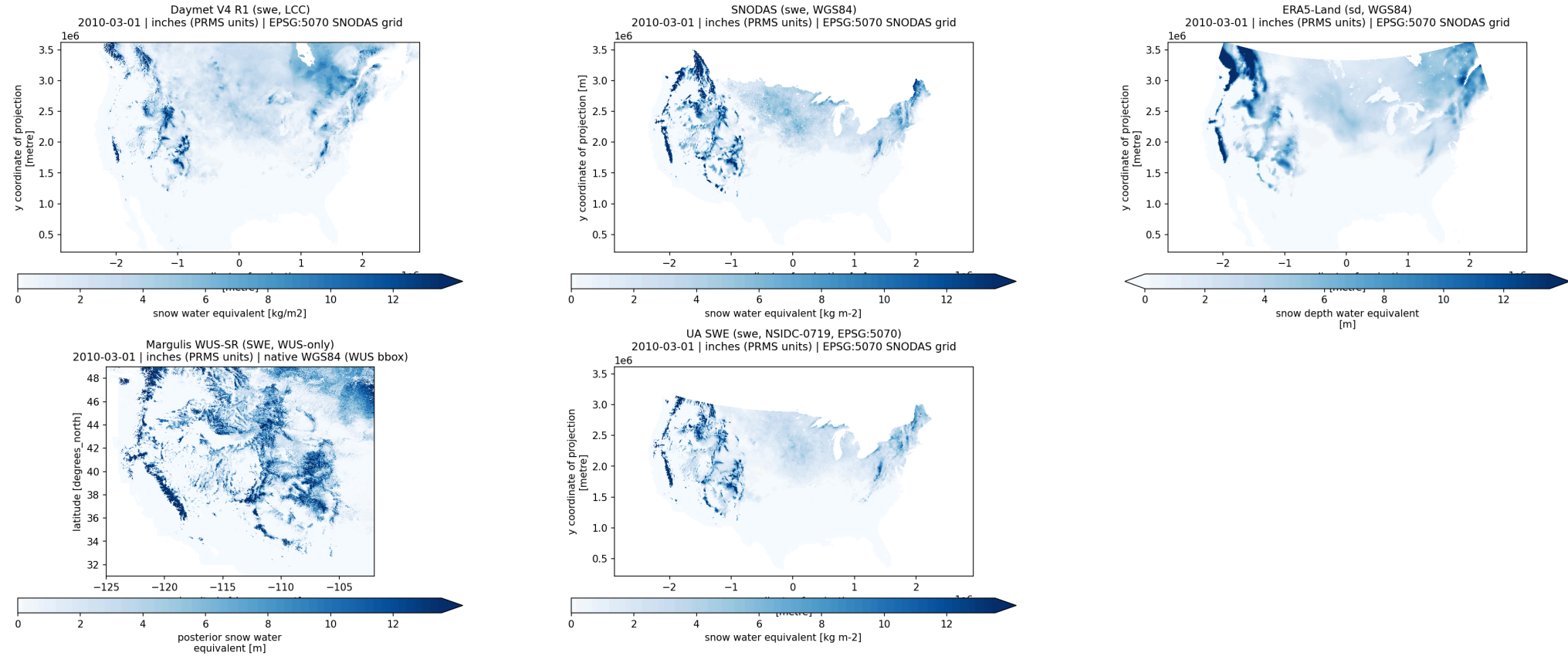
- Same broad pattern across all five: Sierra / Cascades / Rockies / Northern Plains / Northeast.
- UA SWE (4 km) sits between SNODAS/Daymet (1 km, sharp ridges) and ERA5-Land (11 km, smooth).
- Margulis is Western-US-only by design.

Why it matters here

- A new source only helps if it shows the *same physical signal* with *independent* error — visually confirmed.
- 4 km is coarser than our two 1 km sources but far finer than ERA5-Land.

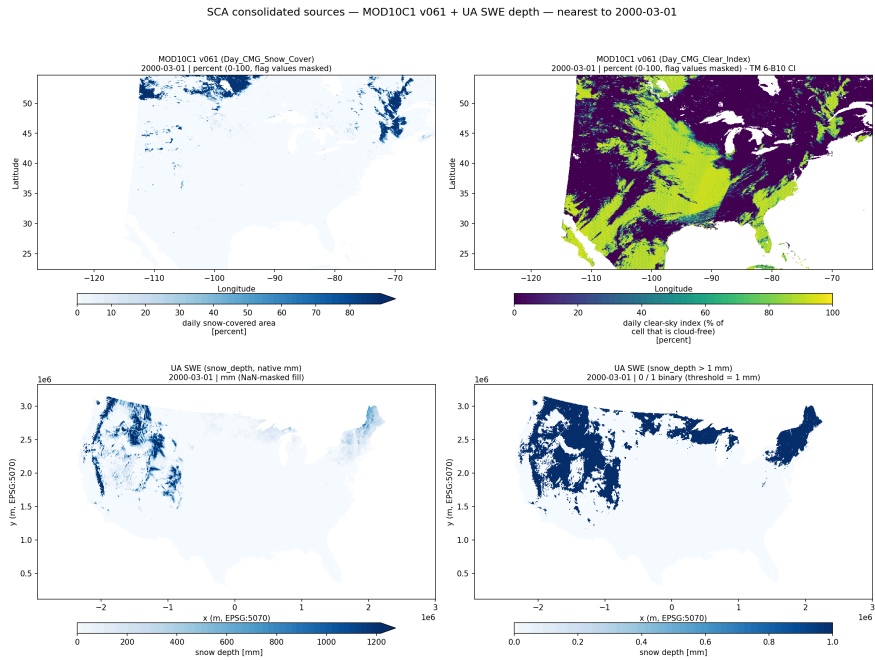
SWE — magnitude check, inches on a shared footprint

SWE consolidated sources — inches, shared SNODAS footprint — 2010-03-01
Daymet/ERA5 reprojected onto SNODAS grid; Margulis kept in native WGS84



All sources → inches (PRMS units), reprojected onto the common SNODAS EPSG:5070 footprint (Margulis kept native WGS84). Colour scale anchored on SNODAS 2nd/98th percentile. The printed table under the notebook cell reports per-source mean + 95th-percentile inches.

Snow-covered area — a depth-derived second source (1 Mar 2000)



Left two: MOD10C1 v061 snow cover + clear-sky index (current single SCA source). Right two: UA SWE snow depth and its `depth > 1 mm` binary — the pixel-level signal the aggregator would area-weight into a per-HRU fractional SCA.

The mechanic

- Pixel `depth > threshold` → 0/1 indicator.
- Area-weighted to HRU = fraction of HRU that is snow-covered.

Why add it to SCA

- MOD10C1 is cloud-gated ($CI \leq 70\%$ drops the day); UA SWE is not.
- Pushes SCA record back to 1982 (MOD10C1 is 2000+).

What it costs, and the open questions

Costs / caveats

- **Resolution.** 4 km vs our 1 km SNODAS/Daymet — coarser, but adds independent info and 20 yr of history.
- **Depth threshold is a free parameter.** Default 1 mm (generous, counts thin snow). Configurable per project; revisit after calibration.
- **Pre-2000 SCA is a point estimate, not a bound.** Before MOD10C1 (2000), the SCA "bound" collapses to UA SWE's single value [v, v]. Honest, but not an interval — will be documented in the target NC.
- **Reprojection at the CONUS boundary.** 4 km NN resample NAD83→EPSG:5070 loses ~1 pixel along high-curvature coast (same trade SNODAS accepts).

Open questions for the room

1. Is a **4 km** third SWE source worth adding alongside two 1 km sources? (We think yes — independent lineage + 1982 record.)
2. Is the **depth-derived SCA** approach sound, and is **1 mm** a sensible starting threshold?
3. Do we want the pre-2000 degenerate SCA interval, or clamp SCA to the MOD10C1 era (2000+)?

The ask — proceed to aggregation + targets?

Done ● — catalog entry, fetch + consolidation (42 WY, 1982-2023), CF-1.6 compliant source grids, these inspection notebooks.

Proposed next, gated on today's decision:

PR	Scope	Unlocks
PR-B	<code>aggregate/ua_swe.py</code> — HRU area-weighting + depth→binary→fraction pre-hook	per-HRU <code>swe</code> , <code>snow_depth</code> , <code>snow_covered_fraction</code>
PR-C	<code>targets/swe.py</code> — add UA SWE as 3rd SWE source	SWE bound back to 1982
PR-D	<code>targets/sca.py</code> — refactor single → multi-source min/max	SCA bound + depth-derived 2nd source

Decision: continue to PR-B → PR-C → PR-D, or hold? If continuing, confirm the 1 mm depth threshold default and the pre-2000 SCA policy.

Per-source detail + magnitude tables: `notebooks/inspect_ua_swe.ipynb` . Cross-source views: `notebooks/consolidated/inspect_consolidated_swe.ipynb` · `inspect_consolidated_snow_covered_area.ipynb` .

Backup / reference

Provenance & reproduction

- **Source:** NSIDC-0719 v1 · DOI `10.5067/0GGPB220EX6A` · <https://nsidc.org/data/nsidc-0719/versions/1>
- **Catalog:** `catalog/sources.yml` → `ua_swe` (units, variables, access block). Source units read from the catalog, never hardcoded.
- **Fetch:** `pixi run nhf-targets fetch ua-swe --project-dir <dir>` (Earthdata-authenticated HTTPS, per-WY NetCDF; SLURM script ships in the project).
- **Consolidation:** `fetch/ua_swe.py::consolidate_water_year_ua_swe` — decode time (`days since 1900-01-01`), rename SWE→ `swe` / DEPTH→ `snow_depth` , mask fill `< 0` , per-day reproject to EPSG:5070 4 km (NN), CF-1.6 metadata via `apply_cf_metadata` .
- **Figures:** `pixi run -e dev render-figures-consolidated` (or the SLURM render job) with `SAVE_FIGURES=1` .

Margulis catalog fix (in PR-A). The `margulis_wus_sr` entry was previously mislabeled "(NSIDC-0719)" — that ID is *this* UA product. Corrected to "(WUS_UCLA_SR)"; Margulis remains the Oregon-scoped 480 m WUS reanalysis.